

Comment on

“Framework for liquid crystal based particle models”

Spacetime Metrics and Connection Singularities

June 22, 2026

Abstract

Ref. (Duda 2021) proposes that topological defect configurations in a liquid-crystal order parameter may define an effective tetrad and emergent Lorentzian spacetime metric representing localized particle states. This Comment establishes that the standard defect-core regularization required to avoid divergent self-energy forces the effective tetrad to be non-invertible at the defect center. Consequently, the emergent metric is degenerate, the inverse tetrad and Weitzenbock connection are singular, and the scalar wave operator is not well-defined across the core. The underlying geometry is therefore singular at the location of the proposed particle state, regardless of the numerical finiteness of the integrated action. This establishes a structural obstruction for emergent-geometry models representing particles as topological defects in the order parameter defining the spacetime metric.

Section 1: Introduction and Mathematical Premise

The framework in Ref. (Duda 2021) utilizes a symmetric matrix field $M_{\mu\nu}$ as the fundamental order parameter. Far from topological defect cores, the field approaches its vacuum configuration, where its eigenvalues λ_a correspond to a chosen background spacetime frame. Near defect cores, the winding of the directors is postulated to represent localized particle excitations, as shown in the model’s structural schema in Figure 1.

To relate the order parameter to spacetime geometry, the frame fields of $M_{\mu\nu}$ define the effective tetrad $e_\mu^a(x)$ and the emergent metric $g_{\mu\nu}(x)$ via the standard mapping:

$$g_{\mu\nu} = \eta_{ab} e_\mu^a e_\nu^b$$

where $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$. Any test fields on this background couple to the geometry through the tetrad e_μ^a and the inverse metric $g^{\mu\nu} = \eta^{ab} e_a^\mu e_b^\nu$.

This Comment addresses the local consistency of this mapping near a topological defect core. The central conflict is stated as follows:

Claim 1. *Let an effective tetrad e_μ^a satisfy the mapping from the local frame of the symmetric tensor $M_{\mu\nu}$ in Ref. (Duda 2021). If core regularization implies that the*

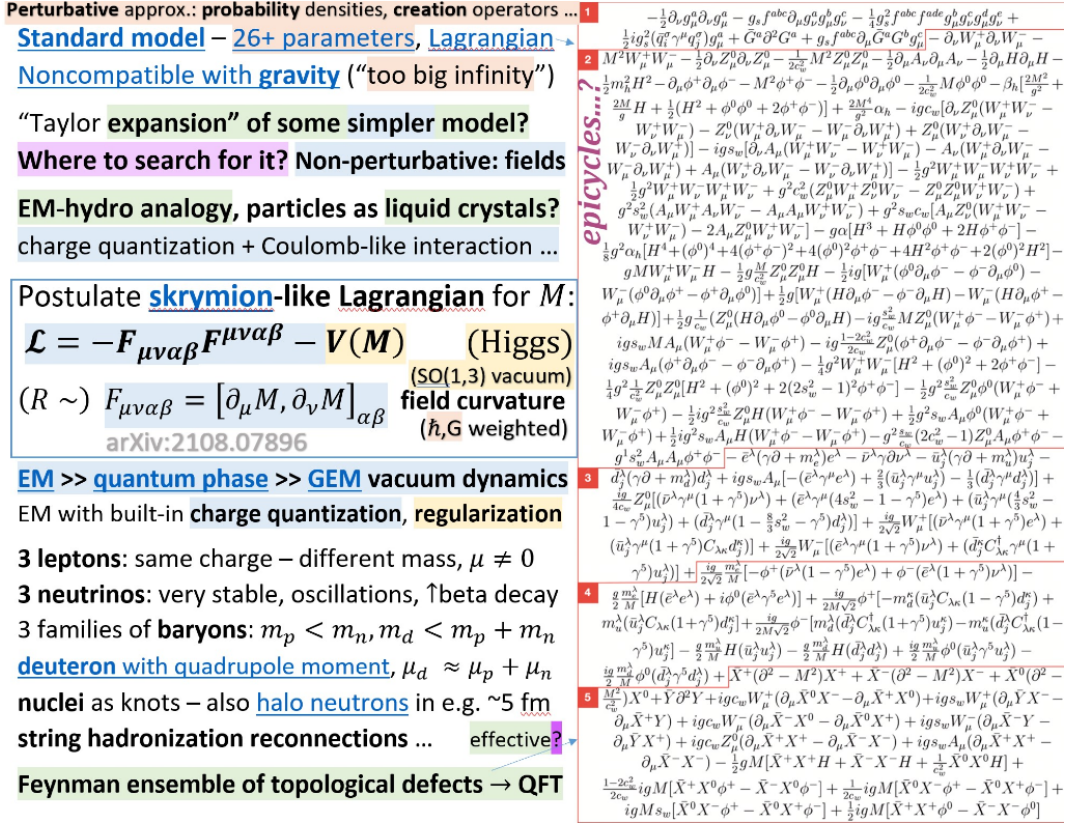


Figure 1: Postulated symmetric matrix field schema and emergent standard model particle correspondence from Ref. (Duda 2021).

eigenvalues are zero at $r = 0$, then the tetrad determinant is zero at $r = 0$, which yields a degenerate metric, a non-invertible tetrad, and a singular Weitzenböck connection at the defect core.

Section 2: Core Regularization and Metric Degeneracy

To model localized particle states with finite field energy, the topological defect core must be regularized. In standard liquid crystal models (e.g., Landau-de Gennes or Skyrme-like theories), this is achieved by letting the order parameter amplitude melt to zero at the origin. Figure 2 illustrates the suppression of the order parameter eigenvalues (isotropic melting) at a nematic-like defect core.

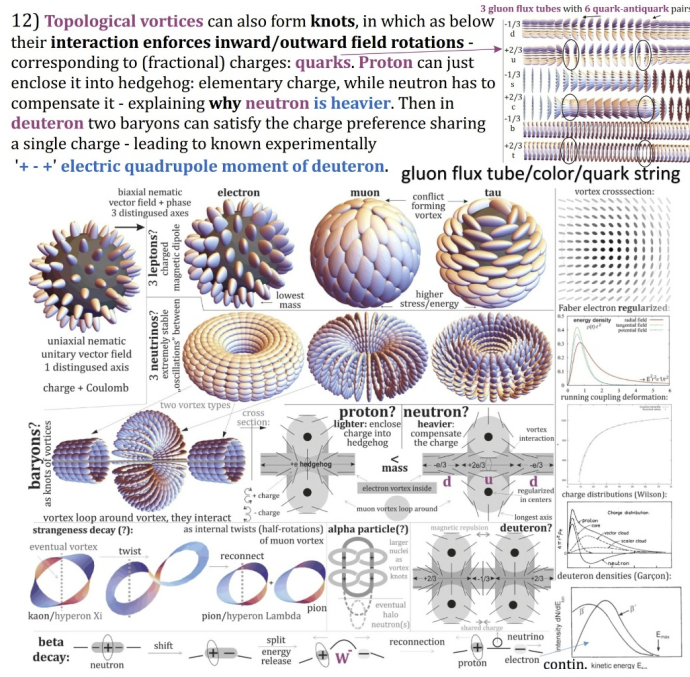


Figure 2: Director field reconstruction and isotropic core melting in a nematic liquid crystal point defect. The shaded central region denotes the core where the order parameter eigenvalues melt to zero to regularize the gradient energy.

In the framework of Ref. (Duda 2021), the local spacetime frame is defined by the symmetric matrix field carrying the topological defect. If the defect core is regularized by vanishing order-parameter amplitudes to maintain finite gradient energy, the amplitudes entering the effective tetrad also vanish at the core.

Consequently, in a spherically symmetric chart $x^\mu = (t, r, \theta, \phi)$ centered on the defect, core regularization requires the frame amplitudes multiplying the local orthonormal directors to vanish as $r \rightarrow 0$. While realistic cores have a finite healing length, the singular limit $r \rightarrow 0$ with power-law vanishing captures the essential geometric obstruction; the finite-core case is analyzed separately in Section 4.3. Regularity of the orthonormal frame on the 2-sphere introduces standard coordinate scaling factors (r and $r \sin \theta$) on the angular components to ensure regularity of the coordinates. This yields the asymptotic radial form for the tetrad:

$$e_t^0 = C_0 r^{\alpha_0}, \quad e_r^1 = C_1 r^{\alpha_1}, \quad e_\theta^2 = C_2 r^{1+\alpha_2}, \quad e_\phi^3 = C_3 r^{1+\alpha_3} \sin \theta$$

where C_a are scaling constants and the boundary exponents satisfy $\alpha_a > 0$ to ensure core regularization. The resulting diagonal metric tensor components are:

$$ds^2 = -C_0^2 r^{2\alpha_0} dt^2 + C_1^2 r^{2\alpha_1} dr^2 + C_2^2 r^{2(1+\alpha_2)} d\theta^2 + C_3^2 r^{2(1+\alpha_3)} \sin^2 \theta d\phi^2$$

The invariant volume element $\sqrt{-g}$ evaluates to:

$$\sqrt{-g} = (C_0 C_1 C_2 C_3) r^{\Sigma \alpha_a + 2} \sin \theta$$

Since the exponents satisfy $\alpha_a > 0$, the volume element vanishes at the origin:

$$\lim_{r \rightarrow 0} \sqrt{-g} = 0$$

This represents a coordinate-invariant rank collapse of the tangent space map. The metric tensor becomes degenerate at $r = 0$. This rank collapse is coordinate-independent and is induced for any local frame field where all eigenvalues melt to zero, regardless of the chosen chart. Consequently, the inverse metric components $g^{\mu\nu}$ diverge component-by-component:

$$g^{tt} = -C_0^{-2} r^{-2\alpha_0}, \quad g^{rr} = C_1^{-2} r^{-2\alpha_1}, \quad g^{\theta\theta} = C_2^{-2} r^{-2(1+\alpha_2)}, \quad g^{\phi\phi} = C_3^{-2} r^{-2(1+\alpha_3)} \csc^2 \theta$$

Thus, the inverse metric tensor does not exist at the core.

Section 3: Teleparallel Formulation and Weitzenböck Connection Singularity

Teleparallel gravity assumes that the tetrad matrix e_μ^a represents a smooth, invertible linear transformation on the tangent bundle. The components of the inverse tetrad e_a^μ satisfy $e_\mu^a e_a^\nu = \delta_\mu^\nu$. Under the radial core scaling, the tetrad determinant is:

$$e = \det(e_\mu^a) = C_0 C_1 C_2 C_3 r^{\alpha_0 + \alpha_1 + \alpha_2 + \alpha_3 + 2} \sin \theta$$

Because the exponents $\alpha_a > 0$ are positive, the determinant vanishes at the core:

$$\lim_{r \rightarrow 0} \det(e_\mu^a) = 0$$

The tetrad matrix is singular at $r = 0$, meaning the inverse tetrad components diverge. The radial inverse component is:

$$e_1^r = C_1^{-1} r^{-\alpha_1}$$

which yields $\lim_{r \rightarrow 0} e_1^r \rightarrow \infty$ for any $\alpha_1 > 0$.

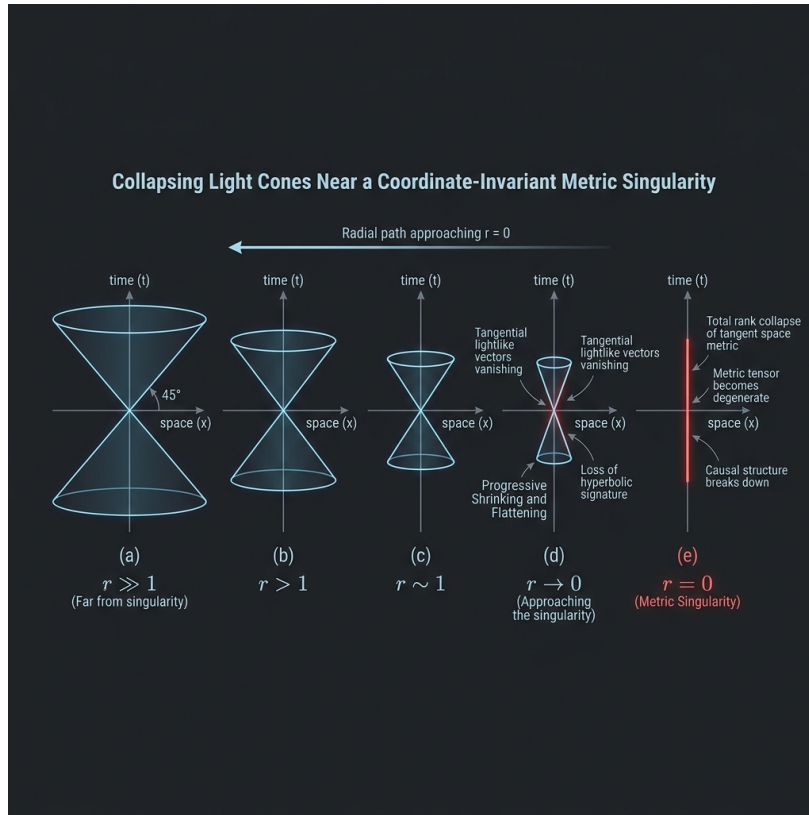


Figure 3: Collapsing light cones illustrating the rank collapse of the tangent space and the break-down of causal structure along a radial path approaching the core origin at $r=0$.

The flat Weitzenbock connection coefficients $\Gamma_{\mu\nu}^\lambda$ are defined by:

$$\Gamma_{\mu\nu}^\lambda = e_a^\lambda \partial_\mu e_\nu^a$$

Substituting the radial components $e_r^1 = C_1 r^{\alpha_1}$ and $e_1^r = C_1^{-1} r^{-\alpha_1}$ into the connection calculation yields:

$$\Gamma_{rr}^r = e_1^r \partial_r e_r^1 = (C_1^{-1} r^{-\alpha_1}) (C_1 \alpha_1 r^{\alpha_1-1}) = \frac{\alpha_1}{r}$$

Therefore, the radial connection component satisfies:

$$\lim_{r \rightarrow 0} \Gamma_{rr}^r = \lim_{r \rightarrow 0} \frac{\alpha_1}{r} \rightarrow \infty$$

This result establishes a structural geometric singularity. The Weitzenbock connection has an irremovable $\mathcal{O}(r^{-1})$ divergence at the defect core, which is independent of the choice of scaling constants C_a or boundary exponents α_a . The connection blow-up is structural: the inverse frame components diverge at the exact rate that the forward frame derivatives decay, rendering parallel transport ill-defined across the core. These connection and torsion scalar divergences are verified symbolically in Ref. (*lcd_geometry_divergences.py* 2026).

Section 4: Corollaries: Wave Propagation and Torsion Invariants

The singular behavior of the inverse metric and tetrad propagates to all associated geometric operators and tensor invariants.

4.1 Divergence of the Scalar Wave Operator

For a massless scalar field $\Phi(x)$, the covariant d'Alembertian is $\square_g \Phi = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \Phi)$. Substituting the radial metric components and volume scaling yields the radial sector:

$$\begin{aligned} \square_g^{(r)} \Phi &= \frac{1}{r \Sigma^{\alpha_a+2}} \partial_r (r \Sigma^{\alpha_a+2} \cdot C_1^{-2} r^{-2\alpha_1} \partial_r \Phi) \\ \square_g^{(r)} \Phi &= C_1^{-2} r^{-2\alpha_1} \partial_r^2 \Phi + C_1^{-2} \left(\frac{\sum \alpha_a - 2\alpha_1 + 2}{r} \right) r^{-2\alpha_1} \partial_r \Phi \\ \square_g^{(r)} \Phi &\sim \mathcal{O}(r^{-2\alpha_1}) \partial_r^2 \Phi + \mathcal{O}(r^{-(2\alpha_1+1)}) \partial_r \Phi \end{aligned}$$

Since the regularization exponent satisfies $\alpha_1 > 0$, the wave operator coefficients diverge near the core, rendering wave propagation ill-defined at the defect location. This implies that the existence of well-defined weak derivatives is excluded under the degenerate metric. For a smooth test function $\psi(x)$ with support containing the origin and non-vanishing gradient, the kinetic energy integral $\int d^4x \sqrt{-g} g^{\mu\nu} \partial_\mu \psi \partial_\nu \psi$ fails to define a Sobolev norm since the inverse metric components diverge.

4.2 Divergence of Torsion Invariants

The torsion tensor $T_{\mu\nu}^\lambda = \Gamma_{\mu\nu}^\lambda - \Gamma_{\nu\mu}^\lambda$ inherits the connection's singular scaling: $T_{\mu\nu}^\lambda \sim \mathcal{O}(r^{-1})$. The coordinate-invariant torsion scalar T is:

$$T = \frac{1}{4}T_{\mu\nu}^\rho T_\rho^{\mu\nu} + \frac{1}{2}T_{\mu\nu}^\rho T_\rho^{\nu\mu} - T_{\mu\rho}^\rho T_\nu^{\nu\mu}$$

Because contracting indices requires multiplying by the divergent components of the inverse metric $g^{\mu\nu} \sim \mathcal{O}(r^{-2\alpha})$, the torsion scalar scales as:

$$T(r) \sim \mathcal{O}(r^{-2-2\alpha})$$

$$\lim_{r \rightarrow 0} T(r) \rightarrow \infty$$

where $\alpha = \max_a \alpha_a$. For homogeneous scaling $\alpha_a = 1$, the torsion scalar diverges as $\mathcal{O}(r^{-4})$, demonstrating a more severe divergence than the connection component. The geometric breakdown of the connection and torsion scalar is verified symbolically in Ref. (*lcd_geometry_divergences.py* 2026).

4.3 Analysis of Loopholes

The Finite-Core Melting Loophole In realistic liquid crystal defect cores, eigenvalues melt to zero over a finite healing length r_0 rather than at a singular point $r = 0$. If a Higgs-like potential $V(M)$ induces isotropic melting over a finite core of radius r_0 , then the tetrad determinant is zero for all $r < r_0$. In this case, rank collapse is induced over a codimension-0 region. The emergent geometry is not well-defined within the core. A finite volume of space is rendered geometrically degenerate, preventing the formulation of covariant differential equations of motion in that region. The codimension-0 rank collapse in this region is verified symbolically in Ref. (*lcd_geometry_divergences.py* 2026).

The Effective Field Theory Loophole If the pseudo-Riemannian metric is interpreted as an effective, long-distance description where short-distance singularities are ignored, the model cannot describe the dynamics of the defect-particles themselves. Since all physical fields couple to the emergent geometry via the tetrad, covariant derivatives inherit the connection divergence. Parallel transport is not well-defined at the core. The geometric pathology is induced in all sectors, preventing the coupling of matter fields.

The Excised Core (Puncture) Loophole The core singularity can be avoided by excising the defect center $r = 0$ from the manifold, treating it as a coordinate puncture. However, this excision alters the physical framework. The defect no longer represents a regular particle excitation on a continuous emergent spacetime backdrop; it constitutes an excised boundary, violating the core objective of deriving particles as smooth configurations.

Section 5: Conclusion

In the liquid-crystal construction proposed by Ref. (Duda 2021), if the defect-carrying order parameter induces the effective tetrad, then core regularization implies that the tetrad is non-invertible at the defect center. The resulting metric degeneracy and $\mathcal{O}(r^{-1})$ connection singularity imply that the putative particle defect is not a smooth, localized excitation of a regular emergent spacetime geometry.

References

Duda, Jarek. 2021. “Framework for Liquid Crystal Based Particle Models.” *arXiv Preprint arXiv:2108.07896*. <https://arxiv.org/abs/2108.07896>.

lcd_geometry_divergences.py. 2026.